

Milestone 1 - Java O-O Program Visualizer/Animator

Team Members:

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Arya Patel arya2023@my.fit.edu
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Faculty Advisor:

David Luginbuhl dluginbuhl@fit.edu

Client:

David Luginbuhl dluginbuhl@fit.edu

Progress of current Milestone:

Task	Completion %	Arya	Aiden	Prottasha	Angelina	To do:
Resolve technical challenges	100%	0%	100%	0%	0%	None
Design memory diagram demo	100%	30%	5%	60%	5%	None
Compare and select the application type	100%	50%	0%	0%	50%	None
Compare and select Collaboration Tools	100%	25%	25%	25%	25%	None
Compare and select technical tools/strategies	100%	0%	100%	0%	0%	Tools for memory instruction animator
Requirement Document	100%	10%	5%	5%	80%	None
Design Document	100%	40%	40%	5%	15%	None
Test Plan	100%	5%	5%	15%	75%	None

Discussion of each accomplished task for the current Milestone:

- Resolve technical challenges: The technical challenge we intended to resolve in this milestone was the lack of technical knowledge. To do so, we researched compilers until we were confident in our ability to apply compiler strategies within the project.
- Design memory diagram demo: To ensure our memory diagrams' visualizations & animations met client standards, we planned to design a primitive, non-programmatic demo to show the client. We used the Canva video editor to create a mockup of our vision

for the memory diagram animations and met with the client to discuss it. The client approved our design, so we plan to move forward with visuals similar to the demo.

- Compare and select the application type: We needed to decide whether to build our software as a web or desktop application. After deliberation, we decided on a desktop application. This is because the compiler strategies we intend to use are easier to implement using the resources a desktop application has access to but a web application does not, such as the ability to run C code natively.
- Compare and select Collaboration Tools: Our group met to discuss how we will collaborate. We decided to use Google Slides for presentation co-development, GitHub for software co-development, Google Docs for documentation co-development, and iMessage for general messaging.
- Compare and select technical tools/strategies: This task required extensive research into common compiler tools and strategies. After conducting the research, it was discovered that programming syntax is often converted into an intermediate language before being compiled into machine code. We have decided that this intermediate language is to be converted into animation instructions.
- Requirement/Design/Test Plan Document: Despite the 8-page length requirement for each document, we completed all three documents on time. We assigned a different person to handle the majority of each document, but all reviewed each one to be on the same page.

Discussion of contribution of each team member to the current Milestone:

- Prottasha Deb: Prottasha put together the majority of the memory diagram demo, meeting with our advisor beforehand for advice. She also wrote part of the Test Plan document and created much of the presentation.
- Angelina Martis: Angelina was responsible for a lot of the documentation work early on. In doing so, she formalized our requirements and test plan for use throughout our project. She also assisted with the decision between which application type to use
- Arya Patel: Arya assisted greatly with both the memory diagram demo and the design document. She also designed a mockup of our GUI for the Software Design Document and as a reference for future milestones. She was also responsible for creating the vast majority of the presentation.
- Aiden Stensrud: Aiden is responsible for the Java parsing aspect of the project. Thus, he was solely responsible for research into compiler tools and strategies, as well as for planning how to apply them to the project. He also wrote a substantial portion of the design documentation.

Plan for the next Milestone:

Task	Arya	Aiden	Prottasha	Angelina
Implement, test, and demo the Java code parser	-	100%	-	-
Implement, test, and demo simple memory diagram generator	20%	-	-	80%
Conceptualize, draft, and demo memory diagram animations	5%	5%	85%	5%
Resolve technical challenge	Memory diagram design for complicated memory states	-	-	-

Discussion of each planned task for the next Milestone:

- Implement, test, and demo the Java code parser: Now that we intend to convert Java code into an intermediate language, we can design an implementation for a simple grammar. As Aiden is specializing in the Java parsing aspect of the project, he will take sole responsibility for that design.
- Implement, test, and demo a simple memory diagram generator: Since we have a mockup of how the memory diagrams should look to the end user, we can begin designing a simple program to generate a memory diagram given a set of instructions. This step will involve designing a syntax for memory diagram instructions (Arya) and designing a program to interpret and execute them (Angelina).
- Conceptualize, draft, and demo memory diagram animations: A demo of how the memory diagrams will shift as instructions execute must also be presented to the client. While Prottasha is designated to do the majority of the work on the demo, each team member will review and provide feedback as needed.
- Resolve technical challenge: Since we plan to digitally animate all relevant low-level information in a given Java program, very complex Java programs could lead to convoluted, crowded memory diagrams, which will require efficient design for the visualizer to both handle and dynamically organize in order to keep the information comprehensible.

Date of meeting with Client during the current milestone:

We met with the client on Tuesday, September 22nd, to demo the memory diagram.

Client feedback on the current milestone:

Faculty Advisor feedback below.

Dates of meeting with Faculty Advisor during the current milestone:

We met with the faculty advisor on Tuesday, September 22nd, to discuss the overall milestone progress.

Faculty Advisor feedback on each task for the current Milestone:

- Resolve technical challenges:
- Design memory diagram demo:
- Compare and select the application type:
- Compare and select Collaboration Tools:
- Compare and select technical tools/strategies:

Faculty Advisor Signature: _____ Date: _____

Evaluation by Faculty Advisor

Faculty Advisor: detach and return this page to Dr. Chan (HC 209) or email the scores to pkc@cs.fit.edu

Score (0-10) for each member: circle a score (or circle two adjacent scores for .25 or write down a real number between 0 and 10)

Prottasha Deb	0	1	2	3	4	5	5.5	6	6.5	7	7.5	8	8.5	9	9.5	10
Angelina Martis	0	1	2	3	4	5	5.5	6	6.5	7	7.5	8	8.5	9	9.5	10
Arya Patel	0	1	2	3	4	5	5.5	6	6.5	7	7.5	8	8.5	9	9.5	10
Aiden Stensrud	0	1	2	3	4	5	5.5	6	6.5	7	7.5	8	8.5	9	9.5	10

Facutly Advisor Signature: _____ Date: _____